

Modeling the Economic Effects of the Section 232 Tariffs on Most Affected Industries

232 Model Release - 2018

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This model accompanies the USITC report, *Economic Impact of Section 232 and 301 Tariffs on U.S. Industries*, Inv. 332-591. The report includes a quantitative analysis of the economic impact of section 232 tariffs on U.S. prices, production and imports. The model inputs (ORANGE-shaded cells) include elasticities that describe responsiveness to price changes, and domestic production and imports data. Data sources are listed in the cell above the data input. The model inputs include the effects of the 232 tariffs. Then, the model simulation estimates what prices and quantities would have looked like if section 232 tariffs were not in place. Outputs are in the GREEN-shaded cells. The white cells are intermediate calculations.

Note: NAICS 336300 is an aggregate of NAICS 336330 and NAICS 336340.

```
In[ ]:= ClearAll[f];  
Clear["Global`*"];
```

1. Policy Changes

Existing Section 232 tariff on aluminum imports

```
In[ ]:= t232a0 = 0.10;
```

Counterfactual Section 232 tariff on aluminum imports

```
In[ ]:= t232a1 = 0.00;
```

Existing Section 232 tariff on steel imports

```
In[ ]:= t232s0 = 0.25;
```

Counterfactual Section 232 tariff on steel imports

```
In[ ]:= t232s1 = 0.00;
```

2. Downstream Industry Definitions

Number of downstream industries

```
In[ ]:= i = 33;
```

Downstream industry NAICS code

```
Inames = {2110, 312110, 3149, 3322, 3323, 3324, 3325, 3326, 3327, 3328,
          3329, 3331, 3332, 3334, 3335, 3339, 3351, 3352, 3353, 3359, 336212, 336214,
          336350, 336370, 336390, 336300, 336411, 3365, 3369, 3372, 3399, 3336, 3366};

In[ ]:= Table[NAICSCodes[t] = Inames[[t]], {t, 1, i}];
```

3. Parameter Inputs

3.1 Elasticity of Substitution

Source: USITC's econometric estimation using the trade cost method in Riker (2020).

Riker, David. "A Trade Cost Approach to Estimating the Elasticity of Substitution." *Economics Working Paper Series 2020-07-D*, U.S. International Trade Commission, July 2020. https://www.usitc.gov/publications/332/working_papers/ecwp_2017-07-d.pdf.

Steel

```
In[ ]:= sigmas = 1.493;
```

Aluminum

```
In[ ]:= sigmaa = 6.046;
```

Downstream industries

```
In[ ]:= sigmad = {5.42, 1.67, 6.33, 9.54, 4.54, 3.84, 5.24, 5.51, 4.22,
                  5, 5.74, 6.95, 7.80, 9.27, 8.90, 7.02, 5.34, 6.67, 9.68, 7.82, 3.06,
                  4.83, 6.48, 4.02, 4.28, 6.44, 3, 7.12, 7.16, 4.27, 6.78, 8.30, 7.51};
```

3.2 Domestic Supply Elasticities

Source: USITC staff estimate based on recent AD/CVD staff reports and data on domestic capacity utilization.

USITC, *Cut-to-Length Carbon-Quality Steel Plate from India, Indonesia, Italy, Japan, and Korea*, December 2011; USITC, *Hot-Rolled Flat-Rolled Carbon-Quality Steel Products from Brazil, Japan, and Russia*, June 2011; USITC, *Certain Hot-Rolled Steel Flat Products from Australia, Brazil, Japan, Korea, the Netherlands, Turkey, and the United Kingdom*, July 2016; USITC, *Cold-Rolled Steel Flat Products from China and Japan*, July 2016; Bown and Russ, "Biden and Europe Remove Trump's Steel and Aluminum Tariffs, but It's Not Free Trade," November 11, 2021.

Domestic steel

```
In[ ]:= eds = 2.5;
```

Domestic aluminum

```
In[ ]:= eda = 4;
```

3.3 Import Supply Elasticities

Source: USITC staff estimate using pass-through results from chapter 6 econometric analysis.

Steel subject imports

```
ln[ ]:= eis = 15;
```

Steel non-subject imports

```
ln[ ]:= ens = 15;
```

Aluminum subject imports

```
ln[ ]:= eia = 20;
```

Aluminum non-subject imports

```
ln[ ]:= ena = 20;
```

4. Data Inputs

4.1 Shipments of U.S. Domestic Steel to Each Downstream Industry

Source: USITC estimates. Total domestic production data were obtained from the U.S. Census Annual Survey of Manufactures. The use of steel by downstream industry was calculated using the Bureau of Economic Analysis (BEA) 2012 Use Table and 2012 Import Matrix with adjustments by industry analysts as necessary. For more information on the steps taken to calculate these values, see Appendix F of the report.

U.S. Census Bureau, "Annual Survey of Manufactures," accessed October 17, 2022. BEA, "Input-Output Accounts Data," accessed October 17, 2022.

```
ln[ ]:= vds0 = {6 260 833, 0, 604 494, 681 098, 10 478 021, 3 275 186, 296 305, 2 522 014, 4 227 251,
              2 160 615, 3 731 414, 5 436 660, 873 797, 1 169 502, 1 194 127, 4 852 505, 511 241, 848 116,
              1 074 890, 1 166 900, 262 110, 338 206, 1 177 140, 5 078 396, 2 560 913, 1 073 461,
              1 266 321, 737 549, 1 043 886, 1 229 248, 1 009 217, 845 515, 298 532}; (*in thousands*)
```

```
ln[ ]:= pds0 = 1;
```

```
ln[ ]:= vdstot0 = Total[vds0];
```

```
ln[ ]:= Table[qds0[t] = (vds0[[t]] / pds0), {t, 1, i}];
```

```
ln[ ]:= qdstot0 = vdstot0 / pds0;
```

4.2 Shipments of U.S. Domestic Aluminum to Each Downstream Industry

Source: USITC estimates. Total domestic production data were obtained from the U.S. Census Annual Survey of Manufactures. The use of aluminum by downstream industry was calculated using the Bureau of Economic Analysis (BEA) 2012 Use Table and 2012 Import Matrix with adjustments by industry analysts as necessary. For more information on the steps taken to calculate these values,

see Appendix F of the report.

U.S. Census Bureau, “Annual Survey of Manufactures,” accessed October 17, 2022. BEA, “Input-Output Accounts Data,” accessed October 17, 2022.

```
ln[ ]:= vda0 = {0, 7 264 652, 0, 322 897, 2 802 539, 6 462 279, 165 977, 48 324,
            1 462 555, 39 622, 1 828 691, 277 247, 327 890, 755 756, 533 068, 1 136 457,
            383 727, 194 624, 250 206, 764 031, 1 032 040, 181 016, 1 391 280, 2 616 144,
            1 319 258, 552 995, 61 419, 68 952, 608 406, 346 541, 345 102, 889 108, 684 310};
(*in thousands*)
```

```
ln[ ]:= pda0 = 1;
```

```
ln[ ]:= vdatot0 = Total[vda0];
```

```
ln[ ]:= Table[qda0[t] = vda0[[t]] / pda0, {t, 1, i}];
```

```
ln[ ]:= qdatot0 = vdatot0 / pda0;
```

4.3 Shipments of Steel Imports to Each Downstream Industry

Source: USITC estimates. Total import data were obtained from the USITC’s DataWeb. The use of steel by downstream industry was calculated using the Bureau of Economic Analysis (BEA) 2012 Use Table and 2012 Import Matrix with adjustments by industry analysts as necessary. For more information on the steps taken to calculate these values, see Appendix F of the report. USITC, “DataWeb/Census,” accessed September 28, 2022. BEA, “Input-Output Accounts Data,” accessed October 17, 2022.

Share of total steel imports that are subject to the Section 232 tariffs

```
ln[ ]:= shrs = 0.4514;
```

Subject imports

```
ln[ ]:= vis0 = shrs * {756 020, 0, 23 556, 277 990, 2 467 548, 726 065, 83 023, 305 340,
                    908 130, 885 164, 1 446 325, 2 042 354, 208 697, 259 754, 401 562, 1 628 308, 84 261,
                    203 804, 315 339, 263 952, 88 085, 110 538, 410 740, 1 144 759, 709 946, 298 616,
                    408 471, 203 804, 322 113, 303 978, 293 836, 308 296, 115 719}; (*in thousands*)
```

```
ln[ ]:= pis0 = 1;
```

```
ln[ ]:= vistot0 = Total[vis0];
```

```
ln[ ]:= Table[qis0[t] = vis0[[t]] / (pis0 (1 + t232s0)), {t, 1, i}];
```

```
ln[ ]:= qistot0 = vistot0 / (pis0 (1 + t232s0));
```

Non-subject imports

```
ln[ ]:= vns0 = (1 - shrs) * {756 020, 0, 23 556, 277 990, 2 467 548, 726 065, 83 023, 305 340,
                    908 130, 885 164, 1 446 325, 2 042 354, 208 697, 259 754, 401 562, 1 628 308, 84 261,
                    203 804, 315 339, 263 952, 88 085, 110 538, 410 740, 1 144 759, 709 946, 298 616,
                    408 471, 203 804, 322 113, 303 978, 293 836, 308 296, 115 719}; (*in thousands*)
```

```
ln[ ]:= pns0 = 1;
```

```
ln[ ]:= vnstot0 = Total[vns0];

ln[ ]:= Table[qns0[t] = vns0[[t]] / (pns0), {t, 1, i}];

ln[ ]:= qnstot0 = vnstot0 / (pns0);
```

4.4 Shipments of Aluminum Imports to Each Downstream Industry

Source: USITC estimates. Total import data were obtained from the USITC's DataWeb. The use of aluminum by downstream industry was calculated using the Bureau of Economic Analysis (BEA) 2012 Use Table and 2012 Import Matrix with adjustments by industry analysts as necessary. For more information on the steps taken to calculate these values, see Appendix F of the report. USITC, "DataWeb/Census," accessed September 28, 2022. BEA, "Input-Output Accounts Data," accessed October 17, 2022.

Share of total aluminum imports that are subject to the Section 232 tariffs

```
ln[ ]:= shra = 0.5413;
```

Subject imports

```
ln[ ]:= via0 = shra * {0, 1673855, 0, 118182, 651967, 2334442, 31157,
18050, 354101, 6590, 455694, 1633, 72040, 199719, 90047, 330252, 78457,
62501, 55724, 194791, 241746, 195957, 83742, 589724, 365729, 153832,
13079, 0, 274256, 146074, 100090, 31027, 148706}; (*in thousands*)
```

```
ln[ ]:= pia0 = 1;
```

```
ln[ ]:= viatot0 = Total[via0];
```

```
ln[ ]:= Table[qia0[t] = via0[[t]] / (pia0 (1 + t232a0)), {t, 1, i}];
```

```
ln[ ]:= qiatot0 = viatot0 / (pia0 (1 + t232a0));
```

Non-subject imports

```
ln[ ]:= vna0 = (1 - shra) * {0, 1673855, 0, 118182, 651967, 2334442, 31157,
18050, 354101, 6590, 455694, 1633, 72040, 199719, 90047, 330252, 78457,
62501, 55724, 194791, 241746, 195957, 83742, 589724, 365729, 153832,
13079, 0, 274256, 146074, 100090, 31027, 148706}; (*in thousands*)
```

```
ln[ ]:= pna0 = 1;
```

```
ln[ ]:= vnatot0 = Total[vna0];
```

```
ln[ ]:= Table[qna0[t] = vna0[[t]] / (pna0), {t, 1, i}];
```

```
ln[ ]:= qnatot0 = vnatot0 / (pna0);
```

4.5 Downstream data

Source: USITC estimates. Total domestic production data were obtained from the U.S. Census Annual Survey of Manufactures or estimated by the Commission's industry analysts using available production data from industry sources, if not available from the ASM. U.S. imports and exports data were obtained from USITC's DataWeb.

U.S. Census Bureau, "Annual Survey of Manufactures," accessed October 17, 2022. USITC, "DataWeb/Census," accessed September

28, 2022.

Downstream domestic production

```
In[ ]:= vd0 = {367 000 000, 44 179 085, 9 803 194, 9 996 673, 97 709 710, 36 409 625,
  9 129 923, 10 003 275, 72 630 014, 28 269 324, 75 165 988, 92 101 680, 36 273 308,
  44 149 565, 33 371 316, 110 247 635, 14 264 428, 23 383 498, 40 073 183, 51 998 246,
  12 069 497, 18 263 830, 42 689 314, 38 327 693, 66 774 309, 28 044 890, 151 871 578,
  12 729 207, 21 252 778, 28 131 184, 61 717 790, 47 930 754, 36 373 643};
(*in thousands*)
```

Downstream domestic exports

```
In[ ]:= vde0 = {76 028 384, 1 151 640, 1 238 552, 2 687 398, 2 085 459, 3 620 919,
  2 784 809, 1 628 277, 3 990 558, 0, 24 309 870, 25 929 153, 23 697 505, 6 960 006,
  6 001 068, 30 785 693, 1 826 734, 8 104 347, 13 884 459, 22 352 409, 1 508 656,
  1 895 592, 3 355 439, 1 507 559, 13 435 966, 14 916 547, 1 299 456, 2 690 284,
  3 964 225, 1 530 847, 18 969 678, 22 574 300, 2 570 885}; (*in thousands*)
```

```
In[ ]:= Table[vdd0[t] = vd0[[t]] - vde0[[t]], {t, 1, i}]
```

```
Out[ ]:= {290 971 616, 43 027 445, 8 564 642, 7 309 275, 95 624 251, 32 788 706, 6 345 114, 8 374 998, 68 639 456,
  28 269 324, 50 856 118, 66 172 527, 12 575 803, 37 189 559, 27 370 248, 79 461 942, 12 437 694,
  15 279 151, 26 188 724, 29 645 837, 10 560 841, 16 368 238, 39 333 875, 36 820 134, 53 338 343,
  13 128 343, 150 572 122, 10 038 923, 17 288 553, 26 600 337, 42 748 112, 25 356 454, 33 802 758}
```

Downstream imports

```
In[ ]:= vi0 = {144 298 174, 3 705 567, 7 539 730, 9 300 961, 6 282 081, 5 229 118, 8 197 750, 3 454 377,
  5 989 155, 0, 37 758 074, 34 864 376, 27 893 070, 16 559 451, 15 611 660, 56 448 298, 15 970 395,
  31 189 044, 34 522 470, 41 280 502, 2 551 100, 773 696, 8 975 144, 596 413, 38 044 695,
  21 919 414, 12 433 623, 1 475 848, 7 564 993, 8 742 444, 88 805 442, 24 445 312, 2 284 897};
(*in thousands*)
```

```
In[ ]:= pi0 = {1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
  1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1};
```

```
In[ ]:= Table[qi0[t] = vi0[[t]] / (pi0[[t]]), {t, 1, i}] ;
```

4.6 Calculated Cost Shares by Downstream Industry

Cost share of aluminum in production of each downstream product

```
In[ ]:= N[Table[csa[t] = (vda0[[t]] + via0[[t]] + vna0[[t]]) / vd0[[t]], {t, 1, i}]]
```

```
Out[ ]:= {0., 0.202324, 0., 0.0441226, 0.0353548, 0.241604, 0.0215921, 0.00663523, 0.0250125,
  0.0016347, 0.0303912, 0.00302796, 0.0110255, 0.0216418, 0.0186722, 0.0133038, 0.0324012,
  0.010996, 0.00763428, 0.0184395, 0.105538, 0.0206404, 0.0345525, 0.0836436, 0.0252341,
  0.0252034, 0.000490533, 0.00541683, 0.0415316, 0.0175113, 0.00721335, 0.0191972, 0.0229016}
```

Cost share of steel in production of each downstream product

```
In[*]:= N[Table[css[t] = (vds0[[t]] + vis0[[t]] + vns0[[t]]) / vd0[[t]], {t, 1, i}]]
```

```
Out[*]:= {0.0191195, 0., 0.0640659, 0.0959407, 0.13249, 0.109895, 0.0415478, 0.282643, 0.070706, 0.107741, 0.0688841, 0.0812039, 0.0298427, 0.032373, 0.0478162, 0.0587841, 0.0417473, 0.0449856, 0.0346923, 0.0275173, 0.0290149, 0.0245701, 0.0371962, 0.162367, 0.0489838, 0.0489243, 0.0110277, 0.0739522, 0.0642739, 0.0545027, 0.0211131, 0.0240725, 0.0113888}
```

Price of other intermediate inputs

[illegible]

Zeroes matrices

[illegible][illegible][illegible][illegible][illegible][illegible]

```
In[*]:= Table[di[t] = If[vi0[[t]] > 0, 1, 0], {t, 1, i}] (*downstream imports*)
```

```
Out[*]:= {1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1}
```

Subject import market shares

$$In[*] := \text{Table}\left[\begin{array}{c} \text{pims}[t] = \text{If}\left[\text{sdinp}[t] + \text{siinp}[t] + \text{sninp}[t] > 0, \frac{\text{vis0}[t]}{\text{vds0}[t] + \text{vis0}[t] + \text{vns0}[t]}, 0\right], \{t, 1, i\}\right];$$
$$In[^\circ] := \text{Table}\left[\right. \\ \left. \text{pima}[t] = \text{If}\left[\text{adinp}[t] + \text{aiinp}[t] + \text{aninp}[t] > 0, \frac{\text{via0}[t]}{\text{vda0}[t] + \text{via0}[t] + \text{vna0}[t]}, 0\right], \{t, 1, i\}\right];$$
$$ln[*]:= \text{Table}\left[\text{pimd}[t] = \text{If}\left[\text{di}[t] > 0, \frac{\text{vi0}[t]}{\text{vi0}[t] + \text{vdd0}[t]}, 0\right], \{t, 1, i\}\right];$$

5. Parameter Calibration

5.1 Supply parameters

$$\ln[\circ] := \text{ads} = \text{qdstot0 pds0}^{-\text{eds}};$$

$$\ln[\circ] := \text{ais} = \text{qistot0 pis0}^{-\text{eis}};$$

$$\ln[\circ] := \text{ans} = \text{qnstot0 pns0}^{-\text{ens}};$$

$$\ln[\circ] := \text{ada} = \text{qdatot0 pda0}^{-\text{eda}};$$

$$\ln[\circ] := \text{aia} = \text{qiatot0 pia0}^{-\text{eia}};$$

$$\ln[\circ] := \text{ana} = \text{qnatot0 pna0}^{-\text{ena}};$$

5.1 Demand parameters

$$\ln[\circ] := \text{Table}[\text{bia}[t] = \text{If}[\text{vda0}[[t]] > 0, \left(\frac{\text{pda0}}{\text{pia0} (1 + t232a0)} \right)^{1-\text{sigmaa}} \left(\frac{\text{via0}[[t]]}{\text{vda0}[[t]]} \right), 0], \{t, 1, i\}];$$

$$\ln[\circ] := \text{Table}[\text{bna}[t] = \text{If}[\text{vda0}[[t]] > 0, \left(\frac{\text{pda0}}{\text{pna0}} \right)^{1-\text{sigmaa}} \left(\frac{\text{vna0}[[t]]}{\text{vda0}[[t]]} \right), 0], \{t, 1, i\}];$$

$$\ln[\circ] := \text{Table}[\text{za0}[t] = \left(\text{pda0}^{1-\text{sigmaa}} + \text{bia}[t] (\text{pia0} (1 + t232a0))^{1-\text{sigmaa}} + \text{bna}[t] \text{pna0}^{1-\text{sigmaa}} \right)^{\frac{1}{1-\text{sigmaa}}}, \{t, 1, i\}];$$

$$\ln[\circ] := \text{Table}[\text{bis}[t] = \text{If}[\text{vds0}[[t]] > 0, \left(\frac{\text{pds0}}{\text{pis0} (1 + t232s0)} \right)^{1-\text{sigmas}} \left(\frac{\text{vis0}[[t]]}{\text{vds0}[[t]]} \right), 0], \{t, 1, i\}];$$

$$\ln[\circ] := \text{Table}[\text{bns}[t] = \text{If}[\text{vds0}[[t]] > 0, \left(\frac{\text{pds0}}{\text{pns0}} \right)^{1-\text{sigmas}} \left(\frac{\text{vns0}[[t]]}{\text{vds0}[[t]]} \right), 0], \{t, 1, i\}];$$

$$\ln[\circ] := \text{Table}[\text{zs0}[t] = \left(\text{pds0}^{1-\text{sigmas}} + \text{bis}[t] (\text{pis0} (1 + t232s0))^{1-\text{sigmas}} + \text{bns}[t] \text{pns0}^{1-\text{sigmas}} \right)^{\frac{1}{1-\text{sigmas}}}, \{t, 1, i\}];$$

Calibration of cost parameters

$$\ln[\circ] := \text{Table}[\text{pd0}[t] = \text{w0}[[t]] + \text{bs0}[t] \times \text{zs0}[t] + \text{ba0}[t] \times \text{za0}[t], \{t, 1, i\}];$$

$$\ln[\circ] := \text{Table}[\text{qd0}[t] = \text{vdd0}[t] / \text{pd0}[t], \{t, 1, i\}];$$

$$\ln[\circ] := \text{Table}[\text{bi}[t] = \left(\frac{\text{pd0}[t]}{\text{pi0}[[t]]} \right)^{1-\text{sigmad}[[t]]} \left(\frac{\text{vi0}[[t]]}{\text{vdd0}[t]} \right), \{t, 1, i\}];$$

$$\ln[\circ] := \text{Table}[\text{P0}[t] = \left(\text{pd0}[t]^{1-\text{sigmad}[[t]]} + \text{bi}[t] \text{pi0}[[t]]^{1-\text{sigmad}[[t]]} \right)^{\frac{1}{1-\text{sigmad}[[t]]}}, \{t, 1, i\}];$$

ln[]:= Table[Cs[t] = css[t] ==

$$\left(\frac{1}{vd0[t]} \right) \left(\left(pds0 \left(\frac{k0[t] \times bs0[t]}{pds0} \right) \left(\frac{pd0[t]}{p0[t]} \right)^{1-sigmad[t]} \left(\frac{zs0[t]}{pd0[t]} \right) \left(\frac{pds0}{zs0[t]} \right)^{1-sigmas} \right) + \right. \\ \left. \left(pis0 (1 + t232s0) \left(\frac{k0[t] \times bs0[t] \times bis[t]}{pis0 (1 + t232s0)} \right) \left(\frac{pd0[t]}{p0[t]} \right)^{1-sigmad[t]} \right. \right. \\ \left. \left(\frac{zs0[t]}{pd0[t]} \right) \left(\frac{pis0 (1 + t232s0)}{zs0[t]} \right)^{1-sigmas} \right) + \left(pns0 \left(\frac{k0[t] \times bs0[t] \times bns[t]}{pns0} \right) \right. \\ \left. \left. \left(\frac{pd0[t]}{p0[t]} \right)^{1-sigmad[t]} \left(\frac{zs0[t]}{pd0[t]} \right) \left(\frac{pns0}{zs0[t]} \right)^{1-sigmas} \right) \right), \{t, 1, i\};$$

ln[]:= Table[Ca[t] =

$$csa[t] = \left(\frac{1}{vd0[t]} \right) \left(\left(pda0 \left(\frac{k0[t] \times ba0[t]}{pda0} \right) \left(\frac{pd0[t]}{p0[t]} \right)^{1-sigmad[t]} \left(\frac{za0[t]}{pd0[t]} \right) \left(\frac{pda0}{za0[t]} \right)^{1-sigmaa} \right) + \right. \\ \left. \left(pia0 (1 + t232a0) \left(\frac{k0[t] \times ba0[t] \times bia[t]}{pia0 (1 + t232a0)} \right) \left(\frac{pd0[t]}{p0[t]} \right)^{1-sigmad[t]} \right. \right. \\ \left. \left(\frac{za0[t]}{pd0[t]} \right) \left(\frac{pia0 (1 + t232a0)}{za0[t]} \right)^{1-sigmaa} \right) + \left(pna0 \left(\frac{k0[t] \times ba0[t] \times bna[t]}{pna0} \right) \right. \\ \left. \left. \left(\frac{pd0[t]}{p0[t]} \right)^{1-sigmad[t]} \left(\frac{za0[t]}{pd0[t]} \right) \left(\frac{pna0}{za0[t]} \right)^{1-sigmaa} \right) \right), \{t, 1, i\};$$

ln[]:= Table[Ck[t] = k0[t] == p0[t]^{1-sigmad[t]} pd0[t]^{sigmad[t]} qd0[t], {t, 1, i}];

```

In[ ]:= sol = FindRoot[{Cs[1], Ca[1], Cs[2], Ca[2], Cs[3], Ca[3], Cs[4], Ca[4], Cs[5], Ca[5],
Cs[6], Ca[6], Cs[7], Ca[7], Cs[8], Ca[8], Cs[9], Ca[9], Cs[10], Ca[10], Cs[11],
Ca[11], Cs[12], Ca[12], Cs[13], Ca[13], Cs[14], Ca[14], Cs[15], Ca[15], Cs[16],
Ca[16], Cs[17], Ca[17], Cs[18], Ca[18], Cs[19], Ca[19], Cs[20], Ca[20], Cs[21],
Ca[21], Cs[22], Ca[22], Cs[23], Ca[23], Cs[24], Ca[24], Cs[25], Ca[25], Cs[26],
Ca[26], Cs[27], Ca[27], Cs[28], Ca[28], Cs[29], Ca[29], Cs[30], Ca[30], Cs[31],
Ca[31], Cs[32], Ca[32], Cs[33], Ca[33], Ck[1], Ck[2], Ck[3], Ck[4], Ck[5],
Ck[6], Ck[7], Ck[8], Ck[9], Ck[10], Ck[11], Ck[12], Ck[13], Ck[14], Ck[15],
Ck[16], Ck[17], Ck[18], Ck[19], Ck[20], Ck[21], Ck[22], Ck[23], Ck[24],
Ck[25], Ck[26], Ck[27], Ck[28], Ck[29], Ck[30], Ck[31], Ck[32], Ck[33]},
{bs0[1], css[1]}, {ba0[1], csa[1]}, {k0[1], vdd0[1] + vi0[1]}, {bs0[2], css[2]},
{ba0[2], csa[2]}, {k0[2], vdd0[2] + vi0[2]}, {bs0[3], css[3]},
{ba0[3], csa[3]}, {k0[3], vdd0[3] + vi0[3]}, {bs0[4], css[4]},
{ba0[4], csa[4]}, {k0[4], vdd0[4] + vi0[4]}, {bs0[5], css[5]},
{ba0[5], csa[5]}, {k0[5], vdd0[5] + vi0[5]}, {bs0[6], css[6]},
{ba0[6], csa[6]}, {k0[6], vdd0[6] + vi0[6]}, {bs0[7], css[7]},
{ba0[7], csa[7]}, {k0[7], vdd0[7] + vi0[7]}, {bs0[8], css[8]}, {ba0[8], csa[8]},
{k0[8], vdd0[8] + vi0[8]}, {bs0[9], css[9]}, {ba0[9], csa[9]}, {k0[9], vdd0[9] + vi0[9]},
{bs0[10], css[10]}, {ba0[10], csa[10]}, {k0[10], vdd0[10] + vi0[10]},
{bs0[11], css[11]}, {ba0[11], csa[11]}, {k0[11], vdd0[11] + vi0[11]},
{bs0[12], css[12]}, {ba0[12], csa[12]}, {k0[12], vdd0[12] + vi0[12]},
{bs0[13], css[13]}, {ba0[13], csa[13]}, {k0[13], vdd0[13] + vi0[13]},
{bs0[14], css[14]}, {ba0[14], csa[14]}, {k0[14], vdd0[14] + vi0[14]},
{bs0[15], css[15]}, {ba0[15], csa[15]}, {k0[15], vdd0[15] + vi0[15]},
{bs0[16], css[16]}, {ba0[16], csa[16]}, {k0[16], vdd0[16] + vi0[16]},
{bs0[17], css[17]}, {ba0[17], csa[17]}, {k0[17], vdd0[17] + vi0[17]},
{bs0[18], css[18]}, {ba0[18], csa[18]}, {k0[18], vdd0[18] + vi0[18]},
{bs0[19], css[19]}, {ba0[19], csa[19]}, {k0[19], vdd0[19] + vi0[19]},
{bs0[20], css[20]}, {ba0[20], csa[20]}, {k0[20], vdd0[20] + vi0[20]},
{bs0[21], css[21]}, {ba0[21], csa[21]}, {k0[21], vdd0[21] + vi0[21]},
{bs0[22], css[22]}, {ba0[22], csa[22]}, {k0[22], vdd0[22] + vi0[22]},
{bs0[23], css[23]}, {ba0[23], csa[23]}, {k0[23], vdd0[23] + vi0[23]},
{bs0[24], css[24]}, {ba0[24], csa[24]}, {k0[24], vdd0[24] + vi0[24]},
{bs0[25], css[25]}, {ba0[25], csa[25]}, {k0[25], vdd0[25] + vi0[25]},
{bs0[26], css[26]}, {ba0[26], csa[26]}, {k0[26], vdd0[26] + vi0[26]},
{bs0[27], css[27]}, {ba0[27], csa[27]}, {k0[27], vdd0[27] + vi0[27]},
{bs0[28], css[28]}, {ba0[28], csa[28]}, {k0[28], vdd0[28] + vi0[28]},
{bs0[29], css[29]}, {ba0[29], csa[29]}, {k0[29], vdd0[29] + vi0[29]},
{bs0[30], css[30]}, {ba0[30], csa[30]}, {k0[30], vdd0[30] + vi0[30]},
{bs0[31], css[31]}, {ba0[31], csa[31]}, {k0[31], vdd0[31] + vi0[31]},
{bs0[32], css[32]}, {ba0[32], csa[32]}, {k0[32], vdd0[32] + vi0[32]},
{bs0[33], css[33]}, {ba0[33], csa[33]}, {k0[33], vdd0[33] + vi0[33]}}];

```

$\text{In}[*]:= \text{Table}[\text{bs}[t] = \text{bs0}[t] /. \text{sol}, \{t, 1, i\}]$

$\text{Out}[*]= \{0.0311401, 0., 0.0855138, 0.324979, 0.250932, 0.300439, 0.108527, 0.65039, 0.123539, 0.242753, 0.231891, 0.244501, 0.150672, 0.0616826, 0.114224, 0.162979, 0.0713078, 0.116545, 0.0956466, 0.0793902, 0.0705263, 0.0512389, 0.0803395, 0.343144, 0.111052, 0.204291, 0.0198414, 0.171027, 0.156717, 0.0976743, 0.0533686, 0.0931034, 0.02473\}$

$\text{In}[*]:= \text{Table}[\text{ba}[t] = \text{ba0}[t] /. \text{sol}, \{t, 1, i\}]$

$\text{Out}[*]= \{0., 0.27321, 0., 0.0794032, 0.0454495, 0.467773, 0.0353585, 0.0128953, 0.0307423, 0.00189229, 0.055016, 0.00477971, 0.0374985, 0.0287582, 0.0255528, 0.0215728, 0.0421389, 0.0194497, 0.0129985, 0.0367975, 0.148602, 0.0280496, 0.0411405, 0.121852, 0.0365568, 0.0671579, 0.000520104, 0.00763705, 0.0631794, 0.0214929, 0.0114205, 0.03979, 0.0266044\}$

$\text{In}[*]:= \text{Table}[\text{k}[t] = \text{k0}[t] /. \text{sol}, \{t, 1, i\}]$

$\text{Out}[*]= \{4.3527 \times 10^8, 4.6733 \times 10^7, 1.61044 \times 10^7, 1.66102 \times 10^7, 1.01906 \times 10^8, 3.80178 \times 10^7, 1.45429 \times 10^7, 1.18294 \times 10^7, 7.46286 \times 10^7, 2.82693 \times 10^7, 8.86142 \times 10^7, 1.01037 \times 10^8, 4.04689 \times 10^7, 5.3749 \times 10^7, 4.29819 \times 10^7, 1.3591 \times 10^8, 2.84081 \times 10^7, 4.64682 \times 10^7, 6.07112 \times 10^7, 7.09263 \times 10^7, 1.31119 \times 10^7, 1.71419 \times 10^7, 4.8309 \times 10^7, 3.74165 \times 10^7, 9.1383 \times 10^7, 3.50478 \times 10^7, 1.63006 \times 10^8, 1.15148 \times 10^7, 2.48535 \times 10^7, 3.53428 \times 10^7, 1.31554 \times 10^8, 4.98018 \times 10^7, 3.60877 \times 10^7\}$

$\text{In}[*]:= \text{Table}[\text{pd0}[t] = (\text{w0}[t] + \text{bs}[t] \times \text{zs0}[t] + \text{ba}[t] \times \text{za0}[t]), \{t, 1, i\}];$

$\text{In}[*]:= \text{Table}[\text{qd0}[t] = \text{vdd0}[t] / \text{pd0}[t], \{t, 1, i\}];$

$\text{In}[*]:= \text{Table}[\text{bi}[t] = \left(\frac{\text{pd0}[t]}{\text{pi0}[t]} \right)^{1-\text{sigmad}[t]} \left(\frac{\text{vi0}[t]}{\text{vdd0}[t]} \right), \{t, 1, i\}];$

$\text{In}[*]:= \text{Table}[\text{Pindex0}[t] = (\text{pd0}[t]^{1-\text{sigmad}[t]} + \text{bi}[t] \text{pi0}[t]^{1-\text{sigmad}[t]})^{\frac{1}{1-\text{sigmad}[t]}}, \{t, 1, i\}];$

$\text{In}[*]:= \text{Table}[\text{za0}[t] = \text{If}[\text{adinp}[t] + \text{aiinp}[t] + \text{aninp}[t] > 0, (\text{pda0}^{1-\text{sigmaa}} + \text{bia}[t] (\text{pia0} (1 + \text{t232a0}))^{1-\text{sigmaa}} + \text{bna}[t] \text{pna0}^{1-\text{sigmaa}})^{\frac{1}{1-\text{sigmaa}}}, 0], \{t, 1, i\}];$

$\text{In}[*]:= \text{Table}[\text{zs0}[t] = \text{If}[\text{sdinp}[t] + \text{siinp}[t] + \text{sninp}[t] > 0, (\text{pds0}^{1-\text{sigmas}} + \text{bis}[t] (\text{pis0} (1 + \text{t232s0}))^{1-\text{sigmas}} + \text{bns}[t] \text{pns0}^{1-\text{sigmas}})^{\frac{1}{1-\text{sigmas}}}, 0], \{t, 1, i\}];$

6. New Equilibrium

$\text{In}[*]:= \text{w} = \text{w0};$

$\text{In}[*]:= \text{pi} = \text{pi0};$

$\text{In}[*]:= \text{Table}[\text{za}[t] = (\text{pda}^{1-\text{sigmaa}} + \text{bia}[t] (\text{pia} (1 + \text{t232a1}))^{1-\text{sigmaa}} + \text{bna}[t] \text{pna}^{1-\text{sigmaa}})^{\frac{1}{1-\text{sigmaa}}}, \{t, 1, i\}];$

```
ln[ ]:= Table[
  zs[t] = (pds1-sigmas + bis[t] (pis (1 + t232s1))1-sigmas + bns[t] pns1-sigmas) $\frac{1}{1-sigmas}$ , {t, 1, i}];
```

```
ln[ ]:= Table[pd[t] = (w[[t]] + bs[t] × zs[t] + ba[t] × za[t]), {t, 1, i}];
```

```
ln[ ]:= Table[P[t] = (pd[t]1-sigmad[[t]] + bi[t] pi[[t]]1-sigmad[[t]]) $\frac{1}{1-sigmad[[t]]}$ , {t, 1, i}];
```

Equilibrium equations

```
ln[ ]:= Table[qda[t] =  $\left(\frac{k[t] \times ba[t]}{pda}\right) \left(\frac{pd[t]}{P[t]}\right)^{1-sigmad[[t]]} \left(\frac{za[t]}{pd[t]}\right) \left(\frac{pda}{za[t]}\right)^{1-sigmaa}$ , {t, 1, i}];
```

```
ln[ ]:= Eda = ada pdaeda == Sum[qda[t], {t, 1, i}];
```

```
ln[ ]:= Table[qia[t] =
   $\left(\frac{k[t] \times ba[t] \times bia[t]}{pia (1 + t232a1)}\right) \left(\frac{pd[t]}{P[t]}\right)^{1-sigmad[[t]]} \left(\frac{za[t]}{pd[t]}\right) \left(\frac{pia (1 + t232a1)}{za[t]}\right)^{1-sigmaa}$ , {t, 1, i}];
```

```
ln[ ]:= Eia = aia piaeia == Sum[qia[t], {t, 1, i}];
```

```
ln[ ]:= Table[qna[t] =  $\left(\frac{k[t] \times ba[t] \times bna[t]}{pna}\right) \left(\frac{pd[t]}{P[t]}\right)^{1-sigmad[[t]]} \left(\frac{za[t]}{pd[t]}\right) \left(\frac{pna}{za[t]}\right)^{1-sigmaa}$ , {t, 1, i}];
```

```
ln[ ]:= Ena = ana pnaena == Sum[qna[t], {t, 1, i}];
```

```
ln[ ]:= Table[qds[t] =  $\left(\frac{k[t] \times bs[t]}{pds}\right) \left(\frac{pd[t]}{P[t]}\right)^{1-sigmad[[t]]} \left(\frac{zs[t]}{pd[t]}\right) \left(\frac{pds}{zs[t]}\right)^{1-sigmas}$ , {t, 1, i}];
```

```
ln[ ]:= Eds = ads pdseds == Sum[qds[t], {t, 1, i}];
```

```
ln[ ]:= Table[qis[t] =
   $\left(\frac{k[t] \times bs[t] \times bis[t]}{pis (1 + t232s1)}\right) \left(\frac{pd[t]}{P[t]}\right)^{1-sigmad[[t]]} \left(\frac{zs[t]}{pd[t]}\right) \left(\frac{pis (1 + t232s1)}{zs[t]}\right)^{1-sigmas}$ , {t, 1, i}];
```

```
ln[ ]:= Eis = ais piseis == Sum[qis[t], {t, 1, i}];
```

```
ln[ ]:= Table[qns[t] =  $\left(\frac{k[t] \times bs[t] \times bns[t]}{pns}\right) \left(\frac{pd[t]}{P[t]}\right)^{1-sigmad[[t]]} \left(\frac{zs[t]}{pd[t]}\right) \left(\frac{pns}{zs[t]}\right)^{1-sigmas}$ , {t, 1, i}];
```

```
ln[ ]:= Ens = ans pnsens == Sum[qns[t], {t, 1, i}];
```

```
ln[ ]:= eqm = FindRoot[{Eda, Eia, Ena, Eds, Eis, Ens}, {pda, pda0},
  {pia, pia0}, {pna, pna0}, {pds, pds0}, {pis, pis0}, {pns, pns0}]
```

```
Out[ ]:= {pda → 0.989887, pia → 1.01817, pna → 0.995896, pds → 0.991952, pis → 1.0183, pns → 0.997939}
```

```
ln[ ]:= pda1 = pda /. eqm;
```

```
ln[ ]:= pia1 = pia /. eqm;
```

```
ln[ ]:= pna1 = pna /. eqm;
```

```

ln[ ]:= pds1 = pds /. eqm;

ln[ ]:= pis1 = pis /. eqm;

ln[ ]:= pns1 = pns /. eqm;

ln[ ]:= Table[za1[t] = If[adinp[t] + aiinp[t] + aninp[t] > 0,
  (pda11-sigmaa + bia[t] (pia1 (1 + t232a1))1-sigmaa + bna[t] pna11-sigmaa) $\frac{1}{1-sigmaa}$ , 0], {t, 1, i}];

ln[ ]:= Table[zs1[t] = If[sdinp[t] + siinp[t] × sninp[t] > 0,
  (pds11-sigmas + bis[t] (pis1 (1 + t232s1))1-sigmas + bns[t] pns11-sigmas) $\frac{1}{1-sigma}$ , 0], {t, 1, i}];

ln[ ]:= Table[pd1[t] = (w[[t]] + bs[t] × zs1[t] + ba[t] × za1[t]), {t, 1, i}];

ln[ ]:= Table[Pindex1[t] = (pd1[t]1-sigmad[[t]] + bi[t] pi[[t]]1-sigmad[[t]]) $\frac{1}{1-sigmad[[t]]}$ , {t, 1, i}];

ln[ ]:= Table[qd1[t] = k[t] Pindex1[t]sigmad[[t]]-1 pd1[t]-sigmad[[t]], {t, 1, i}];

ln[ ]:= qdatot1 = ada pda1eda;

ln[ ]:= qiatot1 = aia pia1eia;

ln[ ]:= qnatot1 = ana pna1ena;

ln[ ]:= qdstot1 = ads pds1eds;

ln[ ]:= qistot1 = ais pis1eis;

ln[ ]:= qnstot1 = ans pns1ens;

ln[ ]:= Table[qda1[t] = If[adinp[t] > 0,
  ( ( (k[t] × ba[t]) / pda1 ) ( pd1[t] / Pindex1[t] )1-sigmad[[t]] ( za1[t] / pd1[t] ) ( pda1 / za1[t] )1-sigmaa ), 0], {t, 1, i}];

ln[ ]:= Table[qia1[t] = If[aiinp[t] > 0, ( (k[t] × ba[t] × bia[t]) / pia1 (1 + t232a1) )
  ( pd1[t] / Pindex1[t] )1-sigmad[[t]] ( za1[t] / pd1[t] ) ( pia1 (1 + t232a1) / za1[t] )1-sigmaa, 0], {t, 1, i}];

ln[ ]:= Table[qna1[t] = If[aninp[t] > 0,
  ( (k[t] × ba[t] × bna[t]) / pna1 ) ( pd1[t] / Pindex1[t] )1-sigmad[[t]] ( za1[t] / pd1[t] ) ( pna1 / za1[t] )1-sigmaa, 0], {t, 1, i}];

ln[ ]:= Table[vda1[t] = If[adinp[t] > 0,
  pda1 * ( (k[t] × ba[t]) / pda1 ) ( pd1[t] / Pindex1[t] )1-sigmad[[t]] ( za1[t] / pd1[t] ) ( pda1 / za1[t] )1-sigmaa, 0], {t, 1, i}];

```

```

ln[ ]:= Table[via1[t] = If[aiinp[t] > 0, pia1 (1 + t232a1) *  $\left(\frac{k[t] \times ba[t] \times bia[t]}{pia1 (1 + t232a1)}\right)$ 
 $\left(\frac{pd1[t]}{Pindex1[t]}\right)^{1-sigmad[[t]]} \left(\frac{za1[t]}{pd1[t]}\right) \left(\frac{pia1 (1 + t232a1)}{za1[t]}\right)^{1-sigmaa}$ , 0], {t, 1, i}];

ln[ ]:= Table[vna1[t] = If[aninp[t] > 0, pna1 *  $\left(\frac{k[t] \times ba[t] \times bna[t]}{pna1}\right)$ 
 $\left(\frac{pd1[t]}{Pindex1[t]}\right)^{1-sigmad[[t]]} \left(\frac{za1[t]}{pd1[t]}\right) \left(\frac{pna1}{za1[t]}\right)^{1-sigmaa}$ , 0], {t, 1, i}];

ln[ ]:= Table[qds1[t] = If[sdinp[t] > 0,
 $\left(\left(\frac{k[t] \times bs[t]}{pds1}\right) \left(\frac{pd1[t]}{Pindex1[t]}\right)^{1-sigmad[[t]]} \left(\frac{zs1[t]}{pd1[t]}\right) \left(\frac{pds1}{zs1[t]}\right)^{1-sigma}$ ), 0], {t, 1, i}];

ln[ ]:= Table[qis1[t] = If[siinp[t] > 0,  $\left(\frac{k[t] \times bs[t] \times bis[t]}{pis1 (1 + t232s1)}\right)$ 
 $\left(\frac{pd1[t]}{Pindex1[t]}\right)^{1-sigmad[[t]]} \left(\frac{zs1[t]}{pd1[t]}\right) \left(\frac{pis1 (1 + t232s1)}{zs1[t]}\right)^{1-sigma}$ , 0], {t, 1, i}];

ln[ ]:= Table[qns1[t] = If[sninp[t] > 0,
 $\left(\frac{k[t] \times bs[t] \times bns[t]}{pns1}\right) \left(\frac{pd1[t]}{Pindex1[t]}\right)^{1-sigmad[[t]]} \left(\frac{zs1[t]}{pd1[t]}\right) \left(\frac{pns1}{zs1[t]}\right)^{1-sigma}$ , 0], {t, 1, i}];

ln[ ]:= Table[vds1[t] = If[sdinp[t] > 0,
pds1 *  $\left(\left(\frac{k[t] \times bs[t]}{pds1}\right) \left(\frac{pd1[t]}{Pindex1[t]}\right)^{1-sigmad[[t]]} \left(\frac{zs1[t]}{pd1[t]}\right) \left(\frac{pds1}{zs1[t]}\right)^{1-sigma}$ ), 0], {t, 1, i}];

ln[ ]:= Table[vis1[t] = If[siinp[t] > 0, pis1 (1 + t232s1) *  $\left(\frac{k[t] \times bs[t] \times bis[t]}{pis1 (1 + t232s1)}\right)$ 
 $\left(\frac{pd1[t]}{Pindex1[t]}\right)^{1-sigmad[[t]]} \left(\frac{zs1[t]}{pd1[t]}\right) \left(\frac{pis1 (1 + t232s1)}{zs1[t]}\right)^{1-sigma}$ , 0], {t, 1, i}];

ln[ ]:= Table[vns1[t] = If[sninp[t] > 0, pns1 *  $\left(\frac{k[t] \times bs[t] \times bns[t]}{pns1}\right)$ 
 $\left(\frac{pd1[t]}{Pindex1[t]}\right)^{1-sigmad[[t]]} \left(\frac{zs1[t]}{pd1[t]}\right) \left(\frac{pns1}{zs1[t]}\right)^{1-sigma}$ , 0], {t, 1, i}];

```

7. Results

Aluminum

$$ln[\circ] := 100 \left(\frac{pda0 - pda1}{pda1} \right)$$

Out[]:= 1.02165

$$ln[\circ] := 100 \left(\frac{pia0 - pia1}{pia1} \right)$$

Out[]:= -1.78503

$$ln[\circ] := 100 \frac{pia0 (1 + t232a0) - pia1 (1 + t232a1)}{pia1 (1 + t232a1)}$$

Out[]:= 8.03646

$$ln[\circ] := 100 \left(\frac{pna0 - pna1}{pna1} \right)$$

Out[]:= 0.412105

$$ln[\circ] := qdachange = 100 \left(\frac{qdatot0 - qdatot1}{qdatot1} \right)$$

Out[]:= 4.14967

$$ln[\circ] := 100 \left(\frac{qiatot0 - qiatot1}{qiatot1} \right)$$

Out[]:= -30.2485

$$ln[\circ] := 100 \left(\frac{qnatot0 - qnatot1}{qnatot1} \right)$$

Out[]:= 8.5729

Steel

$$ln[\circ] := 100 \left(\frac{pds0 - pds1}{pds1} \right)$$

Out[]:= 0.811298

$$\text{In}[*]:= 100 \left(\frac{\text{pis0} - \text{pis1}}{\text{pis1}} \right)$$

Out[*]= -1.79727

$$\text{In}[*]:= 100 \frac{\text{pis0} (1 + \text{t232s0}) - \text{pis1} (1 + \text{t232s1})}{\text{pis1} (1 + \text{t232s1})}$$

Out[*]= 22.7534

$$\text{In}[*]:= 100 \left(\frac{\text{pns0} - \text{pns1}}{\text{pns1}} \right)$$

Out[*]= 0.206563

$$\text{In}[*]:= \text{qdschange} = 100 \left(\frac{\text{qdstot0} - \text{qdstot1}}{\text{qdstot1}} \right)$$

Out[*]= 2.0406

$$\text{In}[*]:= 100 \left(\frac{\text{qistot0} - \text{qistot1}}{\text{qistot1}} \right)$$

Out[*]= -23.8178

$$\text{In}[*]:= 100 \left(\frac{\text{qnstot0} - \text{qnstot1}}{\text{qnstot1}} \right)$$

Out[*]= 3.14365

Downstream

In[*]:= Inames(*naics codes*)

Out[*]= {2110, 312110, 3149, 3322, 3323, 3324, 3325, 3326, 3327, 3328, 3329,
3331, 3332, 3334, 3335, 3339, 3351, 3352, 3353, 3359, 336212, 336214,
336350, 336370, 336390, 336300, 336411, 3365, 3369, 3372, 3399, 3336, 3366}

$$\text{In}[*]:= \text{Table} \left[\text{pddchangex} [t] = 100 \left(\frac{\text{pd0}[t] - \text{pd1}[t]}{\text{pd1}[t]} \right), \{t, 1, i\} \right]$$

Out[*]= {0.0424686, 0.363962, 0.0835914, 0.565853, 0.401414, 0.848729, 0.214501, 0.615166,
0.224489, 0.364945, 0.413511, 0.367393, 0.270995, 0.139751, 0.212235, 0.281324, 0.161106,
0.206407, 0.169114, 0.176138, 0.312644, 0.150102, 0.170869, 0.564407, 0.225789,
0.386083, 0.033646, 0.261841, 0.341529, 0.187464, 0.104798, 0.185241, 0.0821374}


```
In[ ]:= Table[100  $\left( \frac{\text{Pindex0}[t] - \text{Pindex1}[t]}{\text{Pindex1}[t]} \right)$ , {t, 1, i}]
```

```
Out[ ]:= {0.0283965, 0.335087, 0.0444933, 0.251973, 0.376786, 0.732771,  
0.0937707, 0.436891, 0.206514, 0.364945, 0.238095, 0.24137, 0.0846688,  
0.0968464, 0.135507, 0.164961, 0.0706425, 0.0680879, 0.0732195,  
0.0738418, 0.251896, 0.14334, 0.139223, 0.555461, 0.131929, 0.145395,  
0.03108, 0.228476, 0.238208, 0.141167, 0.0341116, 0.0945848, 0.0769479}
```

```
In[ ]:= Table[qddchangex[t] = 100  $\left( \frac{\text{qd0}[t] - \text{qd1}[t]}{\text{qd1}[t]} \right)$ , {t, 1, i}]
```

```
Out[ ]:= {-0.104582, -0.381849, -0.291391, -3.18215, -0.486268, -1.16505,  
-0.722753, -1.40316, -0.281594, -0.363618, -1.23375, -1.1081, -1.52367,  
-0.492837, -0.813783, -0.975086, -0.551608, -0.984507, -0.995346,  
-0.868974, -0.435991, -0.175691, -0.343287, -0.587953, -0.531429, -1.67699,  
-0.0387632, -0.464113, -0.970821, -0.33786, -0.511714, -0.842366, -0.115794}
```

```
In[ ]:= Table[vddchangex[t] = (pd0[t] × qd0[t] - pd1[t] × qd1[t]), {t, 1, i}]
```

```
Out[ ]:= {-180973., -8296.09, -17855.3, -197758., -83080.1, -107308., -32513.3, -67252.8,  
-39653.5, 0., -423231., -496533., -160065., -132034., -166120., -557338., -48871.2,  
-120135., -218630., -207290., -13187., -4232.88, -68167.2, -9894.11, -164167.,  
-172563., -7725.13, -20469.6, -110065., -40235.1, -174891., -168127., -11412.8}
```

```
In[ ]:= Table[vdachange[t] = If[ vda1[t] > 0, vda0[t] - vda1[t], 0], {t, 1, i}];
```

```
In[ ]:= Table[vdschange[t] = If[ vds1[t] > 0, vds0[t] - vds1[t], 0], {t, 1, i}];
```

```
In[ ]:= domesticchange = Prepend[  
Table[{Inames[t], qddchangex[t], pddchangex[t],  $\frac{100 (\text{pd0}[t] \times \text{qd0}[t] - \text{pd1}[t] \times \text{qd1}[t])}{\text{pd1}[t] \times \text{qd1}[t]}$ ,  
vddchangex[t], vdachange[t], vdschange[t]}, {t, 1, i}],  
{"NAICS", "Percent change domestic quantity", "Percent change domestic price",  
"Percent change in value", "Change in value",  
"Change in domestic aluminum", "Change in domestic steel"}];
```

In[]:= Grid[domesticchange, Frame → All]

Out[]:=

NAICS	Percent change domestic quantity	Percent change domestic price	Percent change in value	Change in value	Change in domestic aluminum	Change in domestic steel
2110	-0.104582	0.0424686	-0.0621574	-180 973.	0	133 084.
312 110	-0.381849	0.363962	-0.0192772	-8296.09	363 676.	0
3149	-0.291391	0.0835914	-0.208043	-17 855.3	0	6134.94
3322	-3.18215	0.565853	-2.63431	-197 758.	13 197.1	9780.23
3323	-0.486268	0.401414	-0.0868064	-83 080.1	138 426.	296 707.
3324	-1.16505	0.848729	-0.326205	-107 308.	387 001.	66 975.3
3325	-0.722753	0.214501	-0.509803	-32 513.3	6670.14	8789.29
3326	-1.40316	0.615166	-0.796621	-67 252.8	2853.63	21 183.2
3327	-0.281594	0.224489	-0.0577373	-39 653.5	77 116.8	120 599.
3328	-0.363618	0.364945	0.	0.	1590.01	91 479.6
3329	-1.23375	0.413511	-0.825344	-423 231.	81 600.4	121 099.
3331	-1.1081	0.367393	-0.744773	-496 533.	126.261	178 757.
3332	-1.52367	0.270995	-1.2568	-160 065.	12 272.	16 048.7
3334	-0.492837	0.139751	-0.353775	-132 034.	40 706.3	31 697.1
3335	-0.813783	0.212235	-0.603275	-166 120.	19 302.7	39 406.5
3339	-0.975086	0.281324	-0.696505	-557 338.	60 057.8	152 238.
3351	-0.551608	0.161106	-0.39139	-48 871.2	17 073.2	10 881.3
3352	-0.984507	0.206407	-0.780133	-120 135.	11 038.	20 215.5
3353	-0.995346	0.169114	-0.827914	-218 630.	10 763.7	30 120.
3359	-0.868974	0.176138	-0.694366	-207 290.	37 407.3	27 734.
336 212	-0.435991	0.312644	-0.124711	-13 187.	51 715.3	9607.31
336 214	-0.175691	0.150102	-0.0258536	-4232.88	21 975.5	13 021.1
336 350	-0.343287	0.170869	-0.173004	-68 167.2	28 304.4	45 276.5
336 370	-0.587953	0.564407	-0.0268642	-9894.11	123 867.	134 371.
336 390	-0.531429	0.225789	-0.306839	-164 167.	72 923.6	80 143.9
336 300	-1.67699	0.386083	-1.29738	-172 563.	24 553.3	21 560.7
336 411	-0.0387632	0.033646	-0.00513025	-7725.13	3114.01	50 028.3
3365	-0.464113	0.261841	-0.203487	-20 469.6	379.071	23 512.5
3369	-0.970821	0.341529	-0.632608	-110 065.	43 513.4	30 714.4
3372	-0.33786	0.187464	-0.151029	-40 235.1	25 780.3	37 803.2
3399	-0.511714	0.104798	-0.407452	-174 891.	19 733.8	32 875.8
3336	-0.842366	0.185241	-0.658686	-168 127.	8791.86	29 354.5
3366	-0.115794	0.0821374	-0.0337514	-11 412.8	34 648.1	12 921.8